

Karachi Food Delivery Time Analysis — Auto Analysis

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Executive Summary

This report summarizes the dataset quality, exploratory findings, and model performance for project "Karachi Food Delivery Time Analysis".

Dataset Overview

File	karachi_food_delivery_dataset.csv
Rows	2500
Columns	18
Missing Values	0
Duplicate Rows	0
Cleaned	True

Modeling & Evaluation

Model Type	gradient_boosting
Problem Type	classification
Target Column	late_delivery

Metrics

Metric	Value
accuracy	0.886
precision	0.8841
recall	0.886
f1	0.8802
roc_auc	0.9414
confusion_matrix	{'labels': ['No', 'Yes'], 'matrix': [[362, 13], [44, 81]]}

class_distribution	{'No': 375, 'Yes': 125}
cv_scores	[0.8861623813080124, 0.8522077133855176, 0.8613053613053612, 0.8694995936375247, 0.8742252248997435]
cv_mean	0.8687
baseline	{'baseline_score': 0.6429, 'baseline_metric': 'f1', 'model_score': 0.8802, 'improvement': 0.2373, 'cv_std': 0.0115, 'clearly_beats_baseline': True}
leakage_warnings	[{'column': 'delivery_time_min', 'correlation': 0.9984, 'reason': "Together with 'promised_time_min', this column almost exactly reproduces the target (99.8% agreement via a simple comparison) — the model could 'cheat' by combining the two instead of learning real patterns."}, {'column': 'promised_time_min', 'correlation': 0.9984, 'reason': "Together with 'delivery_time_min', this column almost exactly reproduces the target (99.8% agreement via a simple comparison) — the model could 'cheat' by combining the two instead of learning real patterns."}]
post_outcome_warnings	[{'column': 'tip_pkr', 'reason': "'tip_pkr' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (34% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below."}, {'column': 'customer_rating', 'reason': "'customer_rating' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (46% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below."}]

Feature Importance

Feature	Importance
customer_rating	0.2677
tip_pkr	0.2258
prep_time_min	0.1317
traffic_level_Jam	0.0827
weather_Rainy	0.0481
traffic_level_Low	0.0465
vehicle_type_Bicycle	0.0235
traffic_level_High	0.0202

order_value_pkr	0.0187
rider_experience_months	0.0135
rider_rating	0.0083
items_count	0.0082
vehicle_type_Motorbike	0.0077
weather_Clear	0.0066
distance_km_4.77	0.0032
distance_km_5.77	0.003
distance_km_10.03	0.0028
distance_km_4.65	0.0027
distance_km_4.82	0.0026
distance_km_15.19	0.0026

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Limitations

Model predictions are estimates based on historical data and may contain uncertainty. AI-generated text explains verified analytical results but does not independently calculate metrics. Always validate significant business decisions with domain expertise.